



Intellectual property protection and corporate digital transformation: An empirical analysis from the perspectives of intellectual property protection and digital governance

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ABSTRACT

This paper uses data from Chinese A-share listed companies from 2013 to 2022 as a research sample to explore the relationship between intellectual property protection, digital governance, and corporate digital transformation. The study finds that: intellectual property protection promotes corporate digital transformation; digital governance promotes corporate digital transformation; intellectual property protection enhances corporate innovation, thereby promoting corporate digital transformation; digital governance enhances government transparency, thereby promoting corporate digital transformation; and digital governance plays a positive moderating role in the relationship between intellectual property protection and corporate digital transformation, with this moderating effect exhibiting heterogeneity across eastern, central, and western regions.

1. Introduction

Currently, the Fourth Industrial Revolution, characterized by digitalization, networking, and intelligence, is reshaping the global economic development landscape. As a crucial driving force behind the development of the digital economy, enterprise digital transformation has become a key pathway for enhancing competitiveness, optimizing resource allocation, and achieving high-quality development. With the widespread application of digital technologies such as big data, cloud computing, and artificial intelligence, enterprises need not only to enhance their technological capabilities but also to undertake comprehensive reforms in management models, business processes, and organizational structures (Zhao et al., 2024). However, the smooth advancement of enterprise digital transformation relies heavily on the support of the external institutional environment, particularly the optimization of intellectual property protection and digital governance levels. Intellectual property protection, as a core institutional guarantee for innovation-driven development, directly influences the innovation capacity and technological upgrading processes of enterprises. A robust intellectual property protection system helps enterprises consolidate their technological advantages, reduce innovation risks, and incentivize increased research and development investment, thereby providing intrinsic motivation for digital transformation (Kraus et al., 2022). Furthermore, digital governance, as an important tool for governments to promote the healthy development of the digital economy, not only offers strong support for enterprise digitalization through a comprehensive policy framework and technical support but also creates favorable external conditions for enterprise innovation and transformation by enhancing information

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transparency and optimizing the market environment. Therefore, exploring the impact of intellectual property protection and digital governance on enterprise digital transformation from a dual perspective holds significant theoretical value and practical significance.

Existing literature has extensively examined the relationship between intellectual property protection, digital governance, and enterprise digital transformation. In terms of intellectual property protection, prior studies indicate that it can foster enterprise innovation activities by reducing innovation spillover effects (Xu et al., 2023), enhancing the dynamism of technology markets (Sklenarz et al., 2024), and optimizing resource allocation (Gonçalves et al., 2022). However, research exploring how intellectual property protection further drives digital transformation through enterprise innovation remains relatively scarce. From the perspective of digital governance, it is commonly understood as the government's regulatory and service capacity in the digitalization process (Müller et al., 2024), with its contribution to enterprise digital transformation primarily reflected in policy guidance, infrastructure development, and optimization of public services. However, the mechanisms by which digital governance enhances enterprise digital development through improved government transparency are yet to be fully explored.

Recent studies have focused on the interaction between intellectual property protection and digital governance. Research has found that intellectual property protection and digital governance have synergistic effects in jointly optimizing the innovation environment, reducing transaction costs, and improving resource utilization efficiency, yet empirical studies on how this synergy affects enterprise digital transformation are still insufficient. Additionally, existing literature often focuses on national levels or single industries, lacking in-depth analysis of regional heterogeneity.

This paper uses data from Chinese A-share companies (2013–2022) to explore the drivers of enterprise digital transformation through intellectual property protection and digital governance. The study finds that intellectual property protection boosts enterprise digital transformation, with innovation amplifying this effect. Additionally, digital governance directly drives transformation and enhances it by improving government transparency. Additionally, digital governance positively moderates the link between IP protection and enterprise digital transformation, with notable regional heterogeneity (east, central, west).

The study's contributions lie in three areas: 1) enriching digital transformation research by considering intellectual property protection and digital governance, offering a fresh perspective beyond technical and institutional factors. 2) Revealing how IP protection drives digital transformation via enterprise innovation and clarifying digital governance's indirect impact through transparency. This mechanism analysis not only deepens the understanding of the roles of intellectual property protection and digital governance but also provides a scientific basis for policy formulation. 3) Considering China's regional heterogeneity, it examines digital governance's varied moderating effects, offering empirical backing for tailored regional digital transformation strategies. 4) Based on existing research, this paper systematically analyzes the relationships among intellectual property protection, digital governance, and enterprise digital transformation, thereby broadening the research horizons in related fields and providing theoretical foundations and practical guidance for constructing a more scientific policy support system.

2. Theoretical analysis and research hypotheses

IP protection is crucial for innovation-driven development, aiding digital transformation by: boosting enterprise innovation motivation, reducing technology spillover risks, and increasing innovation returns, thus encouraging R&D investment (Tian et al., 2022). From an economic perspective, the reduction of technology spillover risks aligns with the concept of market failure, where intellectual property rights (IPRs) act as a remedy by providing firms with exclusive rights to their innovations, ensuring that they can capture the full economic benefits of their R&D investments. The core of digital transformation lies in technological innovation; thus, intellectual property protection provides solid technical support for the digital upgrading of enterprises (Zhang & Wang, 2024). This is consistent with endogenous growth theory, which posits that technological innovation is a key driver of long-term economic growth, and strong IP protection ensures that firms have the necessary incentives to invest in innovation, thus fostering growth in the digital economy. Second, intellectual property protection improves the market environment, helping enterprises leverage digital technologies to enhance their competitiveness (Cheng et al., 2023). A more competitive market environment corresponds to perfect competition theory, where strong IP rights reduce barriers to entry, incentivize innovation, and ensure that firms can maintain their competitive advantages in a fairer market, thereby accelerating the digitalization process (Mu et al., 2023). Finally, intellectual property protection can reduce the costs of technological cooperation and transactions, facilitating technology transfer and knowledge sharing. This is in line with transaction cost economics, which suggests that clearer IP protection lowers transaction costs associated with negotiating and enforcing technology licensing and knowledge sharing, thus promoting efficient market exchanges and accelerating the diffusion of digital technologies. These effects collectively propel enterprise digital transformation.

H1. Intellectual property protection promotes the digital transformation of enterprises.

Digital governance involves the government's role in advancing the digital economy through policy guidance, infrastructure development, and regulatory enhancements (Cheng et al., 2024). Effective digital governance can offer enterprises comprehensive and multifaceted support in their digital transformation endeavors. First, digital governance provides a clear direction for enterprise digitalization through policy support and planning. The government fosters a conducive external environment for enterprises undergoing digital transformation by devising comprehensive strategies for digital economic growth and industrial policies that align with technological advancements and market demands (Xing et al., 2021). Second, digital governance can enhance information transparency, reduce information asymmetry, and make it easier for enterprises to access market information, technological resources, and policy support (Niu et al., 2023). Furthermore, government-led digital infrastructure construction (such as 5G networks, data centers, etc.) provides critical technical conditions for enterprises and lowers the entry barriers for digital transformation (Li et al., 2024). In addition, digital governance enhances enterprises' confidence in digital transformation by strengthening cybersecurity and

data regulation.

H2. Regional digital governance promotes the digital transformation of enterprises.

The enhancement of enterprises' innovation capabilities through intellectual property protection is one of the important pathways for promoting digital transformation (Wang & He, 2024). Intellectual property protection provides legal guarantees for enterprises' research and development activities, allowing them to invest resources in innovation with greater peace of mind (Cheng et al., 2023). The improvement of innovation capabilities is directly related to enterprises' technological reserves and application levels in the digital field. For example, when enterprises develop new digital technologies or applications, they rely on patent protection to maintain their technological advantages, thereby promoting digital transformation (Yong & Laing, 2021). Additionally, intellectual property protection can attract external technological cooperation and investment, enabling enterprises to acquire more innovative resources and market opportunities, further driving digital upgrading (Lombardi & Secundo, 2021).

H3. Intellectual property protection promotes the digital transformation of enterprises by enhancing their innovation capabilities.

Government transparency is one of the important manifestations of digital governance, and its role in promoting the digital transformation of enterprises is particularly significant. A highly transparent government can improve the openness of policy information, allowing enterprises to timely understand and utilize relevant digital support policies (Hunjra et al., 2024). At the same time, a transparent regulatory environment helps reduce corruption and rent-seeking behavior, optimizing the market competition environment and providing stable and fair external conditions for the digital development of enterprises (Plekhanov et al., 2023). Furthermore, government transparency can enhance the efficiency of public services, such as rapid approvals and efficient policy implementation, all of which help reduce the costs of digital transformation for enterprises. By increasing transparency, digital governance injects sustained momentum into the digital transformation of enterprises (Zareie et al., 2024).

H4. Regional digital governance promotes the digital transformation of enterprises by enhancing government transparency.

IP protection and digital governance interact significantly, jointly accelerating enterprise digital transformation (Zhai et al., 2022). Digital governance boosts IP protection's role in enhancing enterprise innovation via policy and infrastructure support (Wen et al., 2021). For instance, in regions with high levels of digital governance, enterprises find it easier to utilize digital platforms for technology transactions and knowledge sharing, thereby more effectively converting intellectual property into digital outcomes. Furthermore, there exist disparities in the levels of digital governance among different regions. Specifically, the eastern regions, benefiting from their highly developed digital infrastructure and robust policy support, demonstrate a more pronounced and impactful regulatory effect in this regard (Chen & Hao, 2022), while the central and western regions, due to relatively insufficient resources, show a weaker regulatory effect (Song et al., 2024). Hence, regional digital governance's regulatory role in IP protection and enterprise digital transformation shows notable regional variation.

H5. Regional digital governance can positively moderate the promoting effect of IP protection on the digital transformation of enterprises, and this moderating effect varies across eastern, central, and western regions.

3. Research design

3.1. Data sources

This study uses the A-share listed companies in China from 2013 to 2022 as the initial sample. The data processing includes the following steps: (1) Excluding samples from the financial industry due to the particularity of their financial statements and the inappropriateness of annual report text information as a basis for measuring corporate digital transformation; (2) To avoid reverse causality, excluding companies established in 2013 and later; (3) Excluding samples with missing data on key variables. Ultimately, 18,914 valid sample observations are retained. Corporate digital transformation (indig) is assessed by analyzing listed companies' annual reports using text big data, counting keyword frequencies, and applying logarithmic transformation to create a proxy indicator. Intellectual property protection (property), is sourced from provincial intellectual property statistical yearbooks and the annual reports of the National Intellectual Property Administration. A comprehensive index is calculated using the entropy weight method based on a multidimensional indicator system. Corporate innovation (rd), is derived from the Guotai An database, measured by the ratio of R&D expenditure to operating income, with missing parts supplemented by the Wind database. Digital governance (govern), is based on government open data and third-party evaluation reports, and a comprehensive index is derived through multidimensional standardization and entropy weighting.

3.2. Variable definitions

3.2.1. Dependent variable

Corporate Digital Transformation (indig): This study evaluates the intensity of digital transformation by analyzing text big data from the annual reports of listed companies. It identifies key terms, examines their frequencies (using logarithmic transformation), and uses these as an indicator of the companies' level of digital transformation.

3.2.2. Independent variable

Intellectual Property Protection (property): The comprehensive index of intellectual property is calculated using the entropy weight method, which comprehensively measures the level of intellectual property protection through a multidimensional indicator system. Specifically, the construction of the comprehensive index includes indicators such as intellectual property creation capability (e.g., the number of invention patents per ten thousand people, PCT international patent applications), the strength of intellectual property protection (e.g., the case closure rate of infringement cases, the number of administrative enforcement cases), the level of intellectual property utilization (e.g., the amount of patent pledge financing, the transaction amount of technology contracts), intellectual property management and service capability (e.g., the number of service agencies, the level of special funding support), and the degree of internationalization of intellectual property (e.g., the number of international patent applications, the proportion of intellectual property-intensive goods in exports). The entropy method is used for objective weighting, considering the importance of each indicator to calculate the comprehensive index of intellectual property protection.

3.2.3. Mediating variable

Corporate Innovation (rd): This study measures corporate innovation from the perspective of innovation input, represented by the ratio of R&D expenditure to the previous period's operating income.

Government Transparency (transparent): The government transparency index is approached from five dimensions: policy transparency, information disclosure, public participation, administrative transparency, and ease of information access. It quantifies and evaluates the level of government transparency by combining government open data, third-party evaluations, and public feedback, through standardization and weight assignment. It enhances the robustness of the analysis by factoring in various important dimensions and assigning appropriate significance to each.

3.2.4. Moderating variable

Digital Governance (govern): The digital governance indicator system is developed using the entropy weight method across five dimensions: digital infrastructure (e.g., broadband coverage, number of 5G base stations), data management and security (e.g., data compliance, frequency of security incidents), digital technology application (e.g., use of artificial intelligence and cloud computing), digital governance mechanisms (e.g., transparency of corporate data governance policies, establishment of dedicated departments), and policy and external support (e.g., government subsidies, number of policies promoting digital transformation).

3.2.5. Control variables

This study conducts an in-depth analysis of the impact of intellectual property protection on corporate digital transformation through multiple control variables. Specifically, the control variables include: debt-to-asset ratio (lev), which measures the investment of companies in capital equipment to control for innovation differences in capital-intensive enterprises; investment level (invest), reflecting investment intensity to control for the potential impact of investment changes on digital transformation; current ratio (liquid), describing the level of financial leverage to constrain innovation input; proportion of independent directors (independent), measuring the balance of shareholder power and its influence on digital governance and innovation; Herfindahl index of equity (herfindahl), reflecting the concentration and flexibility of corporate resource allocation; dual role (dual), indicating the state of corporate governance structure to control for its interference with innovation input; total asset growth rate (assetgrowth), measuring the speed of corporate expansion; and operating expense ratio (ofee), reflecting market expectations for corporate growth potential. By setting these control variables, external factors can be effectively minimized, allowing for a more accurate analysis of the relationship between intellectual property protection and corporate digital transformation.

3.3. Model construction

To verify the impact of intellectual property protection and digital governance on corporate digital transformation, this study constructs models (1) and (2), with variable definitions as described above, incorporating a series of control variables. Individual effects $\sum code$ and time effects $\sum year$ are fixed, and a random error term ε is added to ensure the comprehensiveness of the analysis. This method strengthens the validity and robustness of the findings by reducing potential biases arising from both individual and time-related factors.

$$indig_{i,t} = \alpha_0 + \alpha_1 property_{i,t} + \alpha_n control_{i,t} + \sum code + \sum year + \varepsilon_{i,t} \quad (1)$$

$$indig_{i,t} = \theta_0 + \theta_1 govern_{i,t} + \theta_n control_{i,t} + \sum code + \sum year + \varepsilon_{i,t} \quad (2)$$

To delve deeper into the intermediary function of corporate innovation (rd) in the interplay between intellectual property protection and digital transformation, a mediation effect model is meticulously designed and constructed.

$$rd_{i,t} = \mu_0 + \mu_1 property_{i,t} + \mu_n control_{i,t} + \sum code + \sum year + \varepsilon_{i,t} \quad (3)$$

$$indig_{i,t} = \beta_0 + \beta_1 property_{i,t} + \beta_2 rd_{i,t} + \beta_n control_{i,t} + \sum code + \sum year + \varepsilon_{i,t} \quad (4)$$

To explore the mediating role of government transparency (transparent) in the relationship between digital governance and digital transformation, a mediation effect model is constructed.

$$\text{transparent}_{i,t} = \gamma_0 + \gamma_1 \text{govern}_{i,t} + \gamma_n \text{control}_{i,t} + \sum \text{code} + \sum \text{year} + \varepsilon_{i,t} \quad (5)$$

$$\text{indig}_{i,t} = \mu_0 + \mu_1 \text{property}_{i,t} + \mu_2 \text{rd}_{i,t} + \mu_n \text{control}_{i,t} + \sum \text{code} + \sum \text{year} + \varepsilon_{i,t} \quad (6)$$

Finally, to further investigate the impact of digital governance (govern) on the relationship between intellectual property protection and corporate digital transformation, a moderation effect model will be constructed.

$$\text{indig}_{i,t} = \partial_0 + \partial_1 \text{property}_{i,t} + \partial_2 \text{govern}_{i,t} + \partial_3 \text{govern}_{i,t} * \text{property}_{i,t} + \partial_n \text{control}_{i,t} + \sum \text{code} + \sum \text{year} + \varepsilon_{i,t} \quad (7)$$

4. Empirical analysis

4.1. Descriptive statistical analysis

Table 1 shows descriptive stats for main variables. Corporate digital transformation (indig) has a mean of 1.5327 and SD of 0.9734, with wide variation. IP protection (property) has a mean of 0.7899 and limited regional differences. Corporate innovation (rd) is dispersed with high R&D investments. Moderators digital governance (govern) and government transparency (transparent) show low means but significant regional disparities. Control variables like debt-to-asset ratio (lev), independent directors (independent), ownership concentration (herfindahl), and current ratio (liquid) have reasonable distributions. Overall, variable differences provide a solid basis for heterogeneity analysis.

4.2. Benchmark regression analysis

The benchmark regression results in Table 2 indicate that intellectual property protection (property) has a significant promoting effect on corporate digital transformation (indig). In models (1) to (5), the regression coefficients for intellectual property protection are 0.3364, 0.3374, 0.3313, 0.2500, and 0.3378, all of which pass the test at the 1 % significance level. This suggests that for each unit increase in the level of intellectual property protection, the degree of corporate digital transformation will increase by approximately 0.25–0.34 units. This result robustly supports Hypothesis 1, which posits that intellectual property protection can effectively promote corporate digital transformation.

The findings presented in Table 3 underscore the pivotal role of regional digital governance (govern) in fostering enterprise digital transformation (indig), thereby validating Hypothesis 2. Across models (1) to (5), the regression coefficients for digital governance consistently indicate a statistically significant positive impact at the 1 % level, ranging from 0.3224 to 0.3658. This suggests that an incremental unit in the level of digital governance is associated with an approximate increase of between 0.25 and 0.37 units in the extent of enterprise digital transformation. By enhancing digital infrastructure, refining data management and security protocols, and encouraging the adoption of digital technologies, digital governance creates an enabling environment that significantly accelerates and amplifies the digital transformation journey of enterprises.

4.3. Endogeneity test

Table 4 presents the results of the endogeneity analysis, which examines the causal relationship between the core explanatory variable, intellectual property protection (property), and corporate digital transformation (indig) using the instrumental variable approach. The first-stage regression results show that the instrumental variable, L.property, is significantly and positively correlated with intellectual property protection at the 1 % significance level, confirming the appropriateness of the chosen instrument. In the second-stage regression, intellectual property protection significantly influences corporate digital transformation at the 1 % significance level. Additionally, the LM statistic for the endogeneity test is 9061.8954, and the Wald F statistic is 9077.4785, both passing the significance tests. This further validates the instrumental variable and confirms that there are no serious endogeneity concerns in the

Table 1
Descriptive statistics results.

VarName	Obs	Mean	SD	Min	Median	Max
indig	18914	1.5327	0.9734	0.6931	1.0986	6.0210
property	18914	0.7899	0.1130	0.4055	0.8112	0.9375
rd	18914	0.0584	0.2262	0.0000	0.0401	27.3589
transparent	18914	0.1468	3.8574	0.0000	0.0561	317.2884
govern	18914	0.3289	0.1571	0.0257	0.3073	0.7868
lev	18914	0.4079	0.1926	0.0462	0.4007	0.9079
invest	18914	0.0574	0.0604	0.0001	0.0380	0.4083
liquid	18914	2.5235	2.3616	0.2683	1.7673	19.3705
independent	18914	37.9364	5.4873	28.5700	36.3600	60.0000
herfindahl	18914	0.1472	0.1100	0.0106	0.1166	0.5785
dual	18914	0.3274	0.4693	0.0000	0.0000	1.0000
assetgrowth	18914	0.1745	0.3571	−0.3826	0.0931	5.1157
ofee	18914	0.1665	0.1293	0.0100	0.1300	0.7915

Table 2
Basic regression results (1).

VARIABLES	(1)	(2)	(3)	(4)	(5)
	indig				
property	0.3364*** (6.0245)	0.3374*** (5.6421)	0.3313*** (5.5282)	0.2500*** (4.1724)	0.3378*** (5.6279)
lev		−0.4059*** (−11.2466)	−0.4154*** (−8.5075)	−0.3666*** (−7.5544)	−0.2465*** (−4.9558)
invest		−1.1281*** (−9.7579)	−1.1357*** (−9.7734)	−1.1346*** (−9.7560)	−1.3900*** (−10.9133)
liquid			−0.0013 (−0.3301)	−0.0012 (−0.3000)	−0.0052 (−1.3241)
independent			0.0055*** (4.3160)	0.0054*** (4.2423)	0.0052*** (4.1300)
herfindahl				−0.9177*** (−16.0671)	−0.8329*** (−14.4764)
dual				0.1151*** (7.3327)	0.0982*** (6.2546)
assetgrowth					0.1589*** (6.6081)
ofee					0.7053*** (11.7435)
Constant	1.2620*** (28.4919)	1.4940*** (29.2950)	1.2976*** (17.8949)	1.4435*** (19.8765)	1.2060*** (16.2142)
Observations	18,914				
code	YES				
year	YES				
R-squared	0.3002	0.3013	0.3014	0.3028	0.3038

Table 3
Basic regression results (2).

VARIABLES	(1)	(2)	(3)	(4)	(5)
	indig				
govern	0.3224*** (7.4869)	0.3223*** (7.0802)	0.3153*** (6.9203)	0.2536*** (5.5586)	0.3658*** (7.9826)
lev		−0.4180*** (−11.6386)	−0.4156*** (−8.5166)	−0.3671*** (−7.5675)	−0.2412*** (−4.8536)
invest		−1.1097*** (−9.6376)	−1.1142*** (−9.6341)	−1.1200*** (−9.6760)	−1.3963*** (−10.9740)
liquid			0.0001 (0.0254)	−0.0001 (−0.0245)	−0.0040 (−1.0071)
independent			0.0053*** (4.1866)	0.0053*** (4.1323)	0.0050*** (3.9642)
herfindahl				−0.9089*** (−15.9138)	−0.8151*** (−14.1694)
dual				0.1138*** (7.2744)	0.0950*** (6.0667)
assetgrowth					0.1716*** (7.1279)
ofee					0.7424*** (12.2743)
Constant	1.4209*** (92.8212)	1.6583*** (70.8021)	1.4570*** (25.1680)	1.5585*** (26.9598)	1.3457*** (22.6654)
Observations	18,914				
code	YES				
year	YES				
R-squared	0.3003	0.3014	0.3015	0.3029	0.3040

model. These findings indicate a robust causal relationship between intellectual property protection and corporate digital transformation. In conclusion, the endogeneity analysis supports Hypothesis 1, demonstrating that intellectual property protection significantly drives corporate digital transformation, offering solid empirical evidence for policy-making.

4.4. Examination of the innovation mechanism of intellectual property protection

The mechanism analysis results presented in Table 5 support Hypothesis 3, indicating that intellectual property protection indirectly promotes corporate digital transformation by fostering enterprise innovation. Column 1's regression results highlight a

Table 4
Endogeneity test.

	(1)	(2)
	Phase one	Phase two
	property	indig
L.property property	0.2669*** (30.7069)	0.3390*** (3.5394) (3.5394)
Controls	Yes	
LM value	9061.8954	
Wald F value	9077.4785	

Table 5
Mechanism test results (1).

VARIABLES	(1)	(2)
	rd	indig
property	0.0037*** (3.2064)	0.2899** (2.1047)
rd		0.0684** (2.2568)
lev	−0.0070 (−1.0110)	0.0753 (1.0192)
invest	0.0626*** (3.1041)	0.0871 (0.6598)
liquid	0.0036* (1.7157)	−0.0042 (−0.8192)
independent	−0.0001 (−0.8564)	−0.0032* (−1.9391)
herfindahl	−0.0163* (−1.8889)	−0.2803* (−1.8682)
dual	−0.0002 (−0.2284)	−0.0565*** (−2.9892)
assetgrowth	−0.0020 (−0.5703)	0.0050 (0.2433)
ofee	0.1788*** (14.2639)	−0.4497*** (−4.2370)
Constant	0.0191 (0.9139)	1.9999*** (14.9403)
Observations	18,914	
code	YES	
year	YES	
R-squared	0.6934	0.7013

significant positive impact of intellectual property protection on enterprise innovation, with a coefficient of 0.0037 at the 1 % significance level. Column 2 reveals that, when considering enterprise innovation as a mediator, intellectual property protection still has a direct significant effect on corporate digital transformation at the 5 % significance level, with a coefficient of 0.2899, while enterprise innovation also plays a role, with a coefficient of 0.0684, indicating partial mediation. In summary, intellectual property protection not only directly drives corporate digital transformation but also indirectly facilitates the transformation process by enhancing enterprise innovation capabilities, thereby validating the establishment of Hypothesis 3.

4.5. Examination of the transparency transmission mechanism in digital governance

The mechanism analysis for Hypothesis 4 employs “transparency” as a mediator to examine its intermediary role in the relationship between digital governance and corporate digital transformation. The regression outcomes presented in Table 6 offer key insights. Specifically, the results indicate that digital governance (govern) exerts a statistically significant positive influence on transparency (transparent), with a coefficient of 0.3027 that reaches significance at the 5 % level. This finding underscores the capacity of effective digital governance to markedly elevate corporate transparency. Moving to the second column, the impact of transparency on digital transformation (indig) is also notable, exhibiting a coefficient of 0.0034 that is significant at the 1 % level. This further underscores the pivotal role transparency plays in driving corporate digital transformation. Meanwhile, the coefficient for digital governance is 0.3668, indicating that corporate governance indirectly promotes digital transformation by improving transparency. Additionally, the R-squared values for the models are 0.3006 and 0.4040, demonstrating strong explanatory power. In summary, the mechanism test supports Hypothesis 4, which posits that digital governance promotes corporate digital transformation by enhancing corporate

Table 6
Mechanism test results (2).

VARIABLES	(1)	(2)
	transparent	indig
govern	0.3027** (2.5198)	0.3668*** (8.0046)
transparent		0.0034*** (5.3186)
lev	0.5222 (1.3300)	−0.2394*** (−4.8179)
invest	1.0286* (1.8455)	−1.3928*** (−10.9486)
liquid	0.0839 (1.4741)	−0.0037 (−0.9338)
independent	−0.0054** (−2.1828)	0.0050*** (3.9495)
herfindahl	−0.2102*** (−3.2914)	−0.8158*** (−14.1808)
dual	−0.0690 (−0.9650)	0.0948*** (6.0520)
assetgrowth	0.0306 (0.6447)	0.1717*** (7.1320)
ofee	1.9509** (2.4897)	0.7491*** (12.3729)
Constant	−0.5096 (−1.4225)	1.3440*** (22.6356)
Observations	18,914	
code	YES	
year	YES	
R-squared	0.3006	0.4040

transparency.

4.6. Moderating and heterogeneity analysis of digital governance

The results of the moderating effect analysis, as detailed in Table 7, reveal several important findings. Firstly, digital governance (govern) has a statistically significant positive impact on corporate digital transformation (indig), with a coefficient of 0.3195. This underscores the direct and beneficial role that effective digital governance plays in advancing corporate digital transformation. Secondly, intellectual property protection (property) also demonstrates a notable positive effect, with a coefficient of 0.1623, highlighting its contribution to fostering corporate digital transformation. More importantly, the interaction term between digital governance and intellectual property protection (property*govern) exhibits a coefficient of 1.3187 that is significant at the 1 % level. This finding suggests that digital governance amplifies the positive influence of intellectual property protection on corporate digital transformation, indicating a synergistic effect where both factors combined have a more pronounced impact on driving digital transformation within corporations.

The heterogeneity analysis presented in Table 8 provides support for Hypothesis 5, illuminating variations in the moderating role of digital governance on the relationship between intellectual property protection and digital transformation across different regions. Specifically, in the eastern region, the interaction term between intellectual property protection and digital governance is statistically significant and positive, with a coefficient of 0.1313. This finding suggests that regions with more advanced digital governance systems are better equipped to harness the potential of intellectual property protection in facilitating digital transformation. In contrast, the interaction effect between intellectual property protection and digital governance in the central and western regions is not significant. This may be attributed to constraints posed by the levels of economic development and resource allocation in these areas, which may limit their ability to effectively leverage these factors for digital transformation. Consequently, this research indicates that the moderating effect of digital governance is more pronounced and applicable in promoting digital transformation in economically developed regions, such as the eastern part of the country.

In summary, the overall results support Hypothesis 5, which posits that digital governance can positively moderate the relationship between intellectual property protection and corporate digital transformation, and this moderating effect exhibits significant regional heterogeneity, with the eastern region showing the most pronounced effects, while the central and western regions are not significant.

5. Conclusions and recommendations

This paper explores the impact and mechanisms of intellectual property protection and digital governance on corporate digital transformation using data from Chinese A-share listed companies from 2013 to 2022. The study yields significant conclusions. The research results indicate that both intellectual property protection and digital governance significantly promote corporate digital transformation. Specifically, intellectual property protection provides institutional guarantees for corporate innovation activities,

Table 7
Results of moderating effect analysis.

VARIABLES	(1)	(2)
	indig	
property	0.1623** (2.4895)	0.4869*** (3.5350)
govern	0.3195*** (6.4128)	1.3890*** (3.3567)
property*govern		1.3187*** (2.5878)
lev	−0.2393*** (−4.8133)	−0.2329*** (−4.6795)
invest	−1.4161*** (−11.0976)	−1.4049*** (−10.9977)
liquid	−0.0046 (−1.1618)	−0.0042 (−1.0749)
independent	0.0050*** (3.9473)	0.0051*** (4.0052)
herfindahl	−0.8123*** (−14.1153)	−0.8101*** (−14.0861)
dual	0.0924*** (5.8787)	0.0927*** (5.9011)
assetgrowth	0.1711*** (7.0957)	0.1703*** (7.0546)
ofee	0.7485*** (12.3664)	0.7546*** (12.4722)
Constant	1.2350*** (16.6098)	0.9708*** (7.9344)
Observations	18,914	
code	YES	
year	YES	
R-squared	0.3040	0.3040

Table 8
Results of heterogeneity in moderating effects.

VARIABLES	(1)	(2)	(3)
	Eastern region	Central region	Western region
property	0.1831*** (3.7708)	0.2414 (0.5788)	−0.1331 (−0.2474)
govern	0.3180 (0.4961)	1.5906 (1.2692)	−0.1690 (−0.1106)
c.property#c.govern	0.1313*** (3.1691)	−1.1640 (−0.6026)	0.9191 (0.3887)
lev	−0.2069*** (−3.4575)	−0.0939 (−0.7190)	−0.5166*** (−4.4344)
invest	−1.4347*** (−9.5908)	−0.7930** (−2.0816)	−1.3398*** (−4.3052)
liquid	−0.0033 (−0.7308)	−0.0103 (−0.8122)	−0.0171* (−1.8290)
independent	0.0077*** (5.0387)	0.0062* (1.8032)	−0.0100*** (−3.8362)
herfindahl	−0.9140*** (−13.1352)	−0.4357*** (−2.8566)	−0.5596*** (−4.3981)
dual	0.0620*** (3.4619)	0.0892* (1.7488)	0.2774*** (6.5773)
assetgrowth	0.1521*** (5.5499)	0.1745** (2.4671)	0.2088*** (2.8900)
ofee	0.8415*** (11.7115)	0.3059* (1.9524)	0.5036*** (3.3171)
Constant	1.4795*** (7.2435)	0.8976*** (2.7828)	1.9919*** (5.2414)
Observations	14,211	2007	2696
code	YES		
year	YES		
R-squared	0.3040	0.3026	0.3057

stimulates research and development investment and innovative capabilities, thereby advancing the process of corporate digital transformation. This suggests that, in the context of rapid development of the digital economy, the improvement of the intellectual property protection system not only helps to incentivize technological innovation but also provides critical support for corporate transformation and upgrading. Furthermore, the role of digital governance in corporate digital transformation has also been validated. The study shows that digital governance enhances government transparency, creating a more open, fair, and efficient market environment for enterprises, thus significantly facilitating the realization of corporate digital transformation. This result indicates that the widespread application of digital technologies not only changes traditional governance methods but also enhances market efficiency by improving information symmetry and transparency, creating favorable conditions for corporate digital development. Mechanism analysis further reveals that the roles of intellectual property protection and digital governance in corporate digital transformation have interactive effects. Digital governance plays a positive moderating role in the relationship between intellectual property protection and corporate digital transformation. This indicates that when the level of digital governance is high, the promoting effect of intellectual property protection on corporate digital transformation is more pronounced. Regional heterogeneity analysis shows that this moderating effect is most significant in the eastern region, while it is relatively weaker in the central and western regions. The eastern region, with its higher level of economic development, improved market mechanisms, and policy support, can better leverage the synergistic effects of intellectual property protection and digital governance, whereas the central and western regions face certain constraints due to limitations in infrastructure and economic environment. Overall, this study empirically verifies the critical roles and mechanisms of intellectual property protection and digital governance in corporate digital transformation, providing important policy insights for promoting corporate digital transformation and high-quality development. In the future, efforts should be made to further strengthen the construction of the intellectual property protection system and the level of digital governance, particularly in the central and western regions, optimizing resource allocation, narrowing regional disparities, and comprehensively promoting corporate digital transformation.

Based on the findings of this study, several policy recommendations can be made to promote corporate digital transformation. First, strengthening the intellectual property protection system is essential. Policymakers should focus on enhancing legal frameworks and enforcement mechanisms to provide a more secure environment for innovation. By doing so, enterprises will be better incentivized to invest in research and development, thereby accelerating digital transformation. Furthermore, the improvement of digital governance mechanisms should be prioritized, with a focus on increasing government transparency and creating a more open, fair, and efficient market environment. This will enable businesses to more effectively leverage digital technologies for growth and transformation.

In particular, attention should be given to the central and western regions of China, where infrastructure limitations and economic conditions pose challenges. Policies aimed at improving the level of digital governance and intellectual property protection in these regions will help narrow regional disparities, optimize resource allocation, and promote more balanced digital development across the country.

However, there are limitations in this study that should be considered. Although the analysis provides valuable insights, it primarily focuses on data from Chinese A-share listed companies, which may not fully represent the situation in other sectors or smaller enterprises. Additionally, while the study emphasizes the positive roles of intellectual property protection and digital governance, other factors, such as industry-specific dynamics or international influences, were not explored in depth. Future research could expand the scope by including more diverse industries, considering international comparisons, and addressing potential external factors that may impact the digital transformation process.

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Data availability

The authors do not have permission to share data.

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